David Cifuentes

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Valid Ontario G Full Driver's License | Bilingual: English (Professional) & Spanish (Native)

PROFESSIONAL SUMMARY

Lead Automation & Manufacturing Engineer with 10+ years of experience in industrial automation, PLC programming, SCADA/HMI systems, VFDs, and control system integration within manufacturing environments. Proven track record managing multi-site engineering projects across 3 facilities with CAPEX budgets exceeding $950K. Experienced in Rockwell (Allen-Bradley), Siemens, Schneider, and B&R platforms with hands-on experience in system commissioning, troubleshooting, and modernization of legacy control systems. JHSC Certified (Part 1 & 2) with strong knowledge of CSA, ESA, and CEC electrical standards.

CORE COMPETENCIES

- Automation & Controls: Vendor-Agnostic PLC Programming (Allen-Bradley, Siemens, Schneider, B&R), HMI (FactoryTalk, WinCC), SCADA, Robotics, VFDs, Servos.
- Industrial Networks & IIoT: EtherNet/IP, DeviceNet, Profinet, Modbus TCP, MQTT, Node-RED, Python.
- Machine Troubleshooting: Electrical Diagnostics, Pneumatics/Hydraulics, Root Cause Analysis.
- Project Engineering: Equipment Commissioning, Budget Control, BOMs, CAPEX.
- Machine Design: SolidWorks, AutoCAD, Electrical and Panel Design.
- Safety & Standards: CSA C22.1 (CEC), Machine Guarding, LOTO, OHSA.

PROFESSIONAL EXPERIENCE

LEAD AUTOMATION & MANUFACTURING ENGINEER — **ALLEGUARD (Formerly Amvic Systems)**

North York, Scarborough & Calgary, ON/AB — 2023 – Present

- Program, troubleshoot, and optimize PLC-controlled manufacturing systems for EPS production lines.
- Lead commissioning and startup activities including PLC validation, I/O checks, and system integration.
- Diagnoses and resolve electrical, pneumatic, and control system faults under real-time production conditions.
- Rewrite and optimize PLC logic for legacy machinery (Hirsch systems), improving stability and reducing downtime.
- Develop HMI interfaces (FactoryTalk, Weintek) to improve operator interaction and system visibility.
- Integrate PLC systems with industrial networks and IIoT platforms using Node-RED, MQTT, and Python.
- Interpret electrical schematics, P&IDs, and I/O lists for system integration and troubleshooting.
- Implement safety interlocks and machine guarding systems in compliance with CSA and OHSA standards.

AUTOMATION ENGINEER — **AMVIC SYSTEMS**

Toronto & Scarborough, ON — 2021 – 2023

- Designed and implemented PLC-based automation upgrades for production equipment.
- Integrated sensors, VFDs, and actuators into automated control systems.
- Developed control logic, I/O configurations, and system documentation.
- Troubleshot mechanical and electrical issues to improve system reliability.
- Improve logic sequences in PLC programs to reduce production errors.

AMVIC SYSTEMS | Toronto, ON

Maintenance Assistant — 2020 – 2021

- Implemented preventive, predictive, and corrective maintenance programs for manufacturing equipment including EPS moulding machines, boilers, and material handling systems.
- Troubleshot and resolved PLC, HMI, VFD, and sensor issues to minimize production downtime across multiple production lines.
- Configured industrial communication networks (Ethernet/IP, Modbus) for equipment integration and data acquisition.

AUTOMATION CONTROLS ENGINEER **INSTRUCONT SAS**

Bogota, Colombia 2017 – 2019

- Programmed PLCs (Allen-Bradley, Siemens) for industrial applications in the oil & gas sector, including control systems for crude oil separators and accumulator units.
- Developed HMI interfaces and control logic for process automation, enabling real-time monitoring and control of pressure, flow, and level variables.
- Led commissioning activities including I/O verification, loop testing, and system startup for field-installed equipment.
- Integrated SCADA solutions using Modbus and Ethernet/IP protocols for remote monitoring and diagnostics.
- Collaborated with multidisciplinary teams to troubleshoot and optimize process control systems under operational conditions.

Project Manager **DSHAYTECH SAS**

Bogota, Colombia 2014 – 2018

- Designed and built custom automation systems including PLC programming and CNC integration.
- Managed full system lifecycle: design, panel wiring, programming, testing, and commissioning.

MAINTENANCE COORDINATOR **FILMTEX SAS**

Bogota, Colombia 2013 – 2014

- Directed maintenance operations leading a team of 120 technical personnel across mechanical, electrical, and instrumentation disciplines.
- Developed preventive maintenance plans and implemented root cause analysis (RCA) programs to reduce unplanned equipment failures.
- Implement new projects using PLCs Schneider with SoMachine Software, and Siemens with Step 7

EDUCATION

Bachelor of Mechatronic Engineering (WES Validated – Canada Equivalent)

University of San Buenaventura | Bogota, Colombia | 2013

Diploma in Industrial Automation Technologies and PLC Applications

University of San Buenaventura, Bogota, Colombia | 2013

CERTIFICATIONS & TRAINING

- JHSC Certification Part 1 & Part 2 (Worksite Safety) | 2025
- Management Co-Chair, Joint Health and Safety Committee (JHSC)
- CSA C22.1 Canadian Electrical Code – Applied Knowledge